ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **The Wonderful World of Science** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** drop objects of different masses from the same height	Test by a controlled experiment.
2	**(d)** plants need light for healthy growth	Without light, plants turn pale.
3	**(a)** The number alone has no meaning without it	Units give meaning.
4	**(a)** the presence of water	Only one variable changes at a time.
5	**(a)** placing the ruler's zero mark at one end and reading straight on	Correct use of the scale.
6	**(b)** keeping the eye directly above the mark	Avoid parallax by looking straight.
7	**(c)** amount of water and sunlight	Other conditions are kept constant.
8	**(d)** listen to heartbeats	A stethoscope amplifies body sounds.
9	**(c)** two things changed at once, so the cause is unclear	Only one variable should change.
10	**(c)** observation, hypothesis, experiment, conclusion	Standard order of the scientific method.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Then the effect can be linked to that one cause (1); It makes results clear and fair (1).	1+1
12	It collects light from far objects (1); It makes faraway objects look bigger and brighter (1).	1+1
13	Prediction: what we expect to happen (1); Observation: what actually happens (1).	1+1
14	No, it is based on appearance only (1); More evidence from observations shows the Earth moves (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Take two sets of identical seeds with same moisture (1); Keep one set in a warm and another in a cool place (1); Compare sprouting after fixed days (1).	1+1+1
16	Use same size and number of sheets (1); Pour same amount of water on each (1); Compare water absorbed (1).	1+1+1
17	Chemicals or flame may cause harm (1); Safety rules must be known first (1); The teacher's guidance prevents accidents (1).	1+1+1
	OR: Popularity is not evidence (1); Claims must be tested (1); Scientists accept only what experiments support (1).	1+1+1
18	It may have rained (1); A water tanker may have spilt (1); More evidence is required (1).	1+1+1
	OR: Extend senses (1); Give accurate values (1); Allow study of very small or very far objects (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Change: length of the string (1); Keep same: bob, angle of release (1); Measure time for 10 swings (1); Repeat and average (1); Record in a table and conclude (1).	1+1+1+1+1
	OR: Add the readings: 480 cm (1); Average = 480 / 4 = 120 cm (1); Repeating reduces random error (1); Readings vary a little (1); Average is more reliable (1).	1+1+1+1+1
20	Form from observation: plants in shade look pale (1); Hypothesis: plants need sunlight (1); Test with plants in light and dark (1); Pale plants in dark support it (1); Repeat results, then accept (1).	1+1+1+1+1
	OR: Goggles protect eyes (1); Tied hair avoids fire (1); No tasting avoids poison (1); Careful handling of glass avoids cuts (1); Teacher's guidance avoids mistakes (1).	1+1+1+1+1

Mistakes to avoid:

- Calling a guess a **conclusion**: the guess before the experiment is a hypothesis; the conclusion comes after the results.

- Changing many things in an experiment at once; only **one** condition should be changed.
- Writing only what you expected to see instead of what you actually observed.
- Believing that one trial is enough; experiments are repeated to be sure.
- Mixing up the instruments: microscope is for tiny things, telescope is for faraway things.
- Saying science facts never change; new evidence can improve our understanding.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **The Wonderful World of Science** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** measure body temperature	It records temperature.
2	**(c)** The ice cube will melt faster on a metal plate.	A prediction is about a future result.
3	**(a)** Plants kept in the dark will turn yellow.	A prediction that can be tested.
4	**(a)** geologist	Geology.
5	**(a)** evidence from observations	Evidence supports conclusions.
6	**(c)** The pitch is wet, so it must have rained.	It involves reasoning beyond what is seen.
7	**(a)** chemist	Chemists study substances.
8	**(b)** repeating it several times	Repetition increases confidence.
9	**(b)** Which song is the sweetest?	It is a matter of opinion.
10	**(a)** scattering of light	C.V. Raman discovered it.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Conditions and the measurements with units (1); Dates and repeated readings (1).	1+1
12	Scientific study of disease led to vaccines (1); They protect people from diseases (1).	1+1
13	Observation: what we see or measure (1); Conclusion: the meaning drawn from observations (1).	1+1
14	It uses scientific principles like those of electricity (1); New scientific discoveries create new devices (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It is not stated that all else was same (1); Fair test needs one change only (1); Repeated trials are needed (1).	1+1+1
16	No, it still gives information (1); It rules out one explanation (1); A new hypothesis can be made (1).	1+1+1
17	Easy to read (1); Patterns can be found (1); Comparison is simple (1).	1+1+1
	OR: Science discovers principles (1); Technology applies them to build devices (1); Example: electromagnetic principles led to electric motors (1).	1+1+1
18	Wrong placement of the zero (1); Reading from an angle (1); Ruler shorter than the table, giving repeated placement error (1).	1+1+1
	OR: To check for errors (1); To make sure it can be repeated (1); To gain confidence in the result (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Observation: ice melts in sun (1); Hypothesis: heat melts ice (1); Experiment: keep ice in sun and shade (1); Result: melts faster in sun (1); Conclusion: heat melts ice (1).	1+1+1+1+1
	OR: Aryabhata: planets and numbers (1); Charaka: study of diseases (1); Raman: light scattering experiments (1); They observed and reasoned (1); They shared their findings in writing (1).	1+1+1+1+1
20	Fair tests (1); Repeating experiments (1); Accurate instruments (1); Honest records (1); Peer checking by others (1).	1+1+1+1+1
	OR: Science depends on evidence (1); Claims can be tested (1); Results can be repeated (1); Pseudoscience relies on belief (1); It resists testing (1).	1+1+1+1+1

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- Mixing up the instruments: microscope is for tiny things, telescope is for faraway things.
- Saying science facts never change; new evidence can improve our understanding.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **The Wonderful World of Science** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** being ready to change views when evidence changes	Open-mindedness is key.
2	**(b)** Using scientific knowledge to solve practical problems	Technology is applied science.
3	**(a)** launching a satellite	A satellite needs both.
4	**(b)** many tests support it	Evidence is what matters.
5	**(d)** telescope	Telescope gathers light from far objects.
6	**(d)** medicine	It is a classical medicine text.
7	**(c)** preventing certain diseases	They train the body's defence.
8	**(d)** accept the result and revise the hypothesis	Honesty with data is essential.
9	**(c)** improved seeds and irrigation methods	Science improves farming.
10	**(d)** We cannot see very tiny bacteria with the eyes alone	Instruments extend our senses.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Descriptions differ from person to person (1); Measurements give exact values in units (1).	1+1
12	If experiments do not support it (1); Example: seeds sprouted without light, so 'seeds need light' is rejected (1).	1+1
13	So others can verify them (1); So others can build on them (1).	1+1
14	To keep other factors from affecting the result (1); To make results trustworthy (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Questions start investigations (1); Example: asking why the sky is blue led to studies of light scattering (1); Answers can raise new questions (1).	1+1+1
16	Aryabhata worked on astronomy using reasoning (1); Raman studied light scattering through experiments (1); Both used careful observation (1).	1+1+1
17	No, science involves activities (1); Observation and experiments are essential (1); Books record knowledge from them (1).	1+1+1
	OR: Numbers need units for meaning (1); Length is measured in metres (1); Time is measured in seconds (1).	1+1+1
18	Observation: ants gather near sugar (1); Hypothesis: ants are attracted by sweet things (1); Test: place sugar and salt separately (1).	1+1+1
	OR: Vaccines (1); Antibiotics (1); X-ray and scans to diagnose problems (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Early: bullock carts (1); Later: trains (1); Now: aeroplanes (1); Communication: letters to phones (1); Science principles made all these possible (1).	1+1+1+1+1
	OR: Eyes see the Sun moving (1); Touch cannot judge exact temperature (1); Ears cannot hear very faint sounds (1); Thermometer measures temperature exactly (1); Microscope extends sight (1).	1+1+1+1+1
20	Earlier belief: Earth is at the centre (1); New evidence showed Earth moves (1); Earlier belief: germs do not exist (1); Microscopes revealed them (1); So science corrects itself (1).	1+1+1+1+1
	OR: Curiosity leads to questions (1); Questions lead to tests (1); Example: Newton's question on a falling apple (1); It led to the idea of gravity (1); New ideas improve life (1).	1+1+1+1+1

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